

qala

Solution Factory Operating System

Solution System Design Document

Solution Domain Architecture & Specification

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1. Executive Summary

qala is a Solution Factory Operating System (SFOS) — a unified, hierarchical platform for designing, building, managing, and operating solutions of every type, at any scale. The Solution System is the primary domain of the platform. The Solution is the central root element around which every subsystem, environment, factory, workflow, and governance instrument is organized.

Every deliverable managed in qala — whether a software application, a physical good, a market-facing product, a delivered service, or a complex platform — is represented as a typed Solution with a consistent structure, governed lifecycle, and traceable value chain. From a solo developer organizing a personal project to a global enterprise managing thousands of products and services, qala provides a single standardized interface and operating model.

qala is itself a solution factory — the root factory — capable of producing tiered, hierarchical solution factories, development environments, and solution models.

Core Principle:

Every solution, regardless of type or scale, is created, maintained, and operated within a standardized, reproducible, and governed environment. The Solution is the root element of the qala model — everything else exists to enable the Solution.

2. Solution System Overview

The Solution System is the top-level domain of the qala platform. It defines the ontology for what a Solution is, how it is structured, how it is classified, and how it relates to every other system in the platform. All factories, environments, tooling, pipelines, and governance frameworks exist to serve the Solution System.

2.1 The Solution as Root Element

The Solution is the central root element of the qala platform. Every object managed in qala either is a Solution, belongs to a Solution, or exists to support a Solution.

Domain	Relationship to Solution	Description
Solution	Root Element	The fundamental unit of value; typed, versioned, composed of components, governed by lifecycle
Solution Component	Part-of Solution	A constituent element of a Solution; carries design, parts, materials, and IDs
Solution Model	Blueprint of Solution	The design, blueprint, and prototype specification that defines a Solution
Solution Testbed	Validates Solution	The environment and test suites used to verify a Solution
Solution Factory	Produces Solution	The coordinated SDE network that manufactures Solutions
Solution Vendor	Supplies to Solution	External or internal providers of parts, components, or services
Solution Orchestration	Executes for Solution	Workflows and tasks that drive Solution creation and operation
Solution Data	Describes Solution	Metadata, features, versioning, and maturity classification
Solution Tooling	Enables Solution	Toolkits, toolsets, toolchains, and tools used to build Solutions
Solution Book	Documents Solution	Charter, notes, parts, vendors, binders, schedules, WBSs, and collections
Solution Package	Distributes Solution	Bundled, versioned, distributable form of a Solution and its artifacts
Solution Registry	Catalogs Solutions	Authoritative platform-wide catalog of all Solutions, Models, Vendors, and Factories
Solution Portfolio	Aggregates Solutions	A governed collection of Solutions managed under a common Factory or organization
Solution Chain / Set / Kit	Composes Solutions	Higher-order groupings linking multiple Solutions into a coherent offering

2.2 Solution Types

qala recognizes six canonical Solution types. Every artifact managed in the platform is classified as one of these types:

Solution Type	Description
Application	A software application with defined processes and user interactions
System	A composition of applications with coordinated behavior and shared infrastructure
Good	A tangible or digital deliverable produced as an output of a process
Product	A market-facing offering combining software, services, and/or goods
Service	A capability or function delivered to a consumer on-demand or continuously
Platform	A substrate on which other Solutions are built, deployed, and operated

3. Solution

A Solution is the fundamental unit of value in qala. It represents a realized answer to a problem, a goal, or an objective. Solutions are typed, versioned, composed of components, governed by lifecycle states, and linked to the factories, environments, and value chains that produced them.

3.1 Solution Data

Every Solution carries a canonical metadata record and a features list. This data persists in the Solution Registry and is versioned alongside the Solution itself.

Solution Metadata

Metadata Field	Type	Description
Unique ID	UUID v4	Globally unique identifier assigned at Solution creation
Name	string	Human-readable Solution name
Version	semver	Semantic version string (MAJOR.MINOR.PATCH)
Maturity	enum	Lifecycle maturity level: SANDBOX DEV NIGHTLY TEST CM
Solution Type	enum	Application System Good Product Service Platform
Owner	ref	User or organization responsible for the Solution
SDE Reference	ref	Pointer to the SDE in which this Solution was created
Factory Reference	ref	Pointer to the Solution Factory that produced this Solution
Created At	datetime	ISO 8601 timestamp of creation
Updated At	datetime	ISO 8601 timestamp of last modification
Tags	string[]	Freeform labels for search, discovery, and classification

Solution Maturity Model

Each Solution progresses through a defined maturity lifecycle. Promotion between stages requires passing configured gate criteria.

Maturity Stage	Code	Description	Gate Criteria
Sandbox	SANDBOX	Exploratory, unconstrained. No stability guarantees.	None — open creation
Development	DEV	Active feature development. Nightly builds may break.	Basic build success
Nightly	NIGHTLY	Automated nightly build and test cycle. Stability improving.	CI green on main branch
Test	TEST	Feature-complete. Full test suites running. Pre-release hardening.	All test suites passing
Control Managed	CM	Release-qualified. Under Configuration Management. Immutable.	CM board approval + sign-off

Solution Features List

A Solution's capabilities are described by a structured features list. Each feature carries a name, a brief summary, and a full feature specification.

Feature Field	Type	Description
Name	string	Short identifier for the feature
Brief	string	One-sentence summary of what the feature does
Feature	text	Full specification: behavior, acceptance criteria, constraints

4. Solution Component

A Solution Component is the first-level decomposition of a Solution. Components carry design, manufacturing, and material information and can be assembled from lower-level Parts. This model is type-agnostic — a Component may represent a software module, a physical sub-assembly, a service capability, or any other discrete unit.

4.1 Component Structure

Component Field	Description
Component ID	Unique identifier for this component within the Solution
Component Name	Human-readable name
Component Type	Classification: module sub-assembly feature capability service
Design / Blueprint	Reference to the design specification or blueprint document
Version	Component-level semantic version string
Owner	Person or team responsible for this component
Solution Reference	Foreign key back to the parent Solution
Parts List	Ordered collection of Solution Parts belonging to this component

4.2 Solution Part

A Part is the lowest-level constituent of a Component. Parts carry manufacturing and material identity sufficient to uniquely identify, source, and reproduce them.

Part Field	Description
Part ID	Unique identifier (platform-assigned UUID)
Part Number	External or internal part number (e.g., manufacturer PN)
Part Name	Human-readable name for the Part
Part Vendor	The supplier, manufacturer, or source of this Part
Part Material	Material classification: ABS plastic steel open-source library SaaS API etc.
Part Design / Blueprint	Reference to the drawing, specification, or schema defining this Part
Part Version	Version of this Part's design or supply revision
Part Tags	Freeform labels for categorization and search

5. Solution Model

The Solution Model defines the formal specification for a Solution before and during its construction. It is the intellectual blueprint layer — containing designs, blueprints, architectures, mockups, and prototypes — that drives the manufacturing process executed by the Solution Factory.

5.1 Model Components

Model Element	Description
Blueprint	Formal structural specification: architecture, interfaces, data contracts, and dependency graph
Design	Visual and functional design artifacts: wireframes, schematics, drawings, and design system references
Architecture	System-level structure: service topology, data flows, integration patterns, and deployment targets
Mockup	Static visual representation of interfaces and user-facing surfaces for review and sign-off
Prototype	A working instantiation of a subset of Solution capabilities for validation and feedback

5.2 Solution Structure Model

The qala Solution Structure Model defines a universal containment hierarchy for decomposing any Solution. This hierarchy is consistent across all Solution types.

System → Application → Process → Component → Interface → Message → Data Structure → Data

Layer	Contains	Message / Interface Type	Notes
System	Applications	—	Top-level organizational boundary
Application	Processes	—	Executable units with defined behavior
Process	Components	—	Logical groupings of functionality
Component	Interfaces	—	Implementation units with explicit contracts
Interface	Messages + Imports/Exports	Inbound / Outbound	Boundary layer defining data contracts
Message	Data Structures	Event (dynamic) / State (static)	Typed message payload
Data Structure	Data fields	—	Typed fields using qala data type system
Data	Primitive values	—	Leaf-level values using canonical data types

5.3 Data Type System

qala defines a canonical set of data types used across all Solution data structures. Custom types may be composed from these primitives.

Primitive Types	Numeric Types	Collection Types	Reference Types
bool	int	array	pointer
string	float	tuple	object
char	double	set	custom
varchar	null	map	—
date	—	—	—

6. Solution Testbed

The Solution Testbed is the verification and validation environment for a Solution. It defines the full battery of tests, test suites, testbed infrastructure, and quality gates that a Solution must pass before advancing through the maturity lifecycle.

6.1 Testbed Architecture

Testbed Element	Description
Test Case	The atomic unit: a single scenario with defined inputs, expected outputs, and pass/fail criteria
Test Suite	A named, ordered collection of related test cases targeting a specific feature, component, or concern
Test Plan	The master document defining scope, schedule, environments, and acceptance criteria for a release
Test Environment	An isolated, automatically provisioned environment matching the target deployment topology
Test Run	A single execution of a test suite against a specific Solution build, producing a dated results record
Defect	A recorded failure: linked to the test case that found it, the build, and the release it blocks
Test Analytics	Aggregated metrics across runs: pass rates, trend analysis, defect density, MTTR

6.2 Test Types

Test Type	Scope	Automation Level
Unit Tests	Individual functions and modules	Fully automated — runs on every commit
Integration Tests	Service-to-service and module boundaries	Fully automated — runs on every build
System Tests	End-to-end Solution behavior	Automated — runs on nightly and release builds
Performance / Benchmark	Throughput, latency, scalability under load	Automated — runs on release candidates
Security / SAST / DAST	Static and dynamic vulnerability scanning	Automated — SAST on commit, DAST on builds
Acceptance Tests (UAT)	Business requirements validation	Semi-automated — human sign-off required
Prototype Tests	POC feasibility and design validation	Manual + AI-assisted feedback analysis
Regression Tests	Prevention of previously resolved defects	Fully automated — runs on every build

7. Solution Factory (SF)

A Solution Factory (SF) is a coordinated, networked collection of Solution Development Environments (SDEs) that produces Solutions in a consistent, repeatable, and scalable way. The factory provides the production infrastructure — hermetic builds, CI/CD pipelines, artifact management, and governance controls — that transforms Solution Models into deployed Solutions.

7.1 Factory Structure

Factory Element	Description
Solution Factory	Top-level coordination entity; owns a collection of networked SDEs
Solution Development Environment (SDE)	The atomic operational unit — see Section 8 for full specification
Solution Network	The interconnected chain of SDEs operated by this Factory, networked and orchestrated
Build System	Hermetic, reproducible build infrastructure for all Solutions produced by the Factory
CI/CD System	Automated integration, testing, and deployment pipelines
Artifact Repository	Versioned storage for all build outputs, binaries, containers, and packages
Solution Registry	Platform-level catalog of all Solutions, SDEs, and produced artifacts within this Factory
Solution Portfolio	The governed collection of Solutions produced and managed by this Factory
Governance Layer	Policies, templates, standards, and approval workflows applied to all factory output

7.2 Factory Hierarchy

qala itself is the Root Solution Factory. It produces tiered, hierarchical child factories that inherit governance, tooling, and pipeline configurations from their parent.

Factory Tier	Scope	Governance
Root Factory (qala)	Operates all child factories; platform-level governance	Full platform governance — immutable policies
Enterprise Factory	Multi-team, multi-product factory with full governance suite	Inherits root + enterprise policy customization
Team Factory	Single-team factory scoped to a product or service line	Inherits enterprise + team-level overrides
Personal Factory	Single-user factory for personal and hobby projects	Lightweight governance; owner is sole approver

7.3 Hermetic Build Environments

All Solution builds within qala are executed in hermetic build environments — fully isolated, reproducible build contexts that guarantee consistent outputs regardless of when or where they are run.

- Fully isolated: no access to external network resources, host file system, or ambient credentials during build
- Reproducible: identical inputs always produce byte-identical outputs
- Environment-as-code: all build environment definitions are version-controlled and immutable at build time
- Containerized: each build executes in a purpose-built, ephemeral container spun up and destroyed per run
- Dependency locking: all dependency versions are pinned and cryptographically verified before execution
- Build attestation: every build produces a signed SLSA provenance record linking output to inputs

7.4 CI/CD/CM Pipeline

Pipeline Stage	Trigger	Actions
Continuous Integration (CI)	Every commit / pull request	Build, unit test, SAST scan, artifact generation, code quality gate
Continuous Delivery (CD)	CI green + merge to main	Integration tests, system tests, DAST scan, staging deployment
Continuous Deployment	CD green + maturity gate	Automated promotion: canary, blue/green, or rolling deployment
Configuration Management (CM)	CM board approval	Immutable release tagging, audit trail, environment lock-down, sign-off

8. Solution Development Environment (SDE)

The Solution Development Environment (SDE) is the atomic operational unit of qala. Every developer, team, or automated process works within an SDE. SDEs are solution-agnostic by design — they define the conditions for creating, maintaining, and operating solutions, not the solutions themselves. One SDE can support a collection of solutions.

8.1 SDE Properties

Property	Description
Deployable	Can be deployed to a platform, cloud environment, or physical machine
Configurable	All parameters, settings, and options are externally definable
Distributable	Can be shared, cloned, replicated, and distributed across teams or sites
Version-controlled	Full history of environment state, including snapshots and rollbacks
Composable	Multiple SDEs can be combined or linked within a Solution Factory
Scalable	Supports single-user and enterprise-scale workloads without model change
Releasable	SDEs can be versioned and distributed as sealed, reproducible environment packages

8.2 SDE Internal Architecture

Solution Configuration

Configuration Element	Description
Solution Version	The version of the Solution this SDE is configured to produce
Solution Component	References to each component, with component-level version pinning
Solution Part	References to each part, with part-level version and vendor specification
Environment Variables	Runtime configuration values, secrets references, and parameter overrides
Configuration Files	Versioned, structured config files (YAML, TOML, JSON, ENV)
Settings / Parameters / Options	All SDE-level settings governing behavior, tooling, and runtime
User Preferences & Profiles	Developer-specific preferences, IDE bindings, and workspace layouts

Solution Model (within SDE)

Model Element	Description
Solution Blueprint	Formal structural specification defining architecture and interfaces
Solution Design	Visual and functional design artifacts linked to this SDE's solution scope
Solution Architecture	System-level topology, service maps, data flow, and integration patterns
Solution Mockup	Static interface representations for stakeholder review
Solution Prototype	Working POC or partial implementation for validation

Solution Environments (within SDE)

Environment	Purpose	Characteristics
Solution Sandbox Environment	Open-ended exploration and experimentation	No stability guarantees; ephemeral; no governance gates
Solution Build / Assembly Environment	Hermetic build and solution assembly	Isolated; reproducible; dependency-locked; attested
Solution Test Environment	Full verification and validation	Isolated clone of production topology; auto-provisioned per run
Solution Release Environment	Pre-release staging and CM sign-off	Production-equivalent; immutable after CM gate; DAST-scanned

Solution Build (within Build Environment)

Build Element	Description
Solution Build	A discrete, versioned output of the build pipeline for a Solution
Solution Build Version	Semantic version of this build output (MAJOR.MINOR.PATCH-BUILD)
Solution Build Maturity	Maturity level at time of build: SANDBOX DEV NIGHTLY TEST CM
Solution Build Number	Monotonically incrementing build sequence number within the version
Solution Assembly	The composed, integrated output of all Components combined into a deliverable unit

Solution Toolbox

Tooling Element	Description
Tool	Atomic unit: a single executable, library, or service performing a specific function
Toolset	A named collection of related Tools serving a common purpose
Toolkit	A curated collection of Toolsets covering a full domain of practice
Toolchain	A linked, ordered sequence of Tools where each tool's output feeds the next
Tool Suite	A governed, platform-approved set of Toolkits for a given Solution type
IDE Integrations	VSCode, JetBrains, and other developer environment plugins and extensions
3rd-Party Integrations	Vendor tool integrations managed, versioned, and qualified within the SDE

8.3 SDE Lifecycle States

State	Trigger	Description
Provisioning	SDE creation request	Environment being assembled from template or definition
Active	Provisioning complete	Fully operational; development and build activities occur
Snapshotted	User or CI action	State captured; environment continues to run
Suspended	Inactivity or admin action	Resources released; state preserved for resumption
Rolled Back	Restore request	Environment restored to a prior snapshot version

State	Trigger	Description
Archived	Retention policy or admin	Moved to long-term storage; retrievable on demand
Terminated	Explicit deletion	Permanently decommissioned; backups retained per policy

8.4 SDE Maturity

Each SDE carries a maturity classification aligned to the Solution it is producing. The SDE maturity gates control which operations are permitted within the environment — for example, only CM-maturity SDEs may produce immutable release artifacts.

8.5 SDE Backup, Recovery & Archiving

Operation	Description
Snapshot	Point-in-time capture of full SDE state; supports rollback and branching
Rollback	Restore SDE to a prior snapshot version; CI/CD pipelines resume immediately
Backup	Scheduled or event-driven backup: full or incremental; stored in versioned secure storage
Clone	Create an independent copy of an SDE for parallel development or benchmarking
Archive	Move inactive or legacy SDEs to long-term storage with retention policy enforcement
Restore	Rehydrate an archived or backed-up SDE to an active state

9. Solution Book

The Solution Book is the complete knowledge and documentation repository for a Solution. It consolidates all records, plans, structures, schedules, and organizational artifacts that govern and describe a Solution from inception through retirement. Every Solution has exactly one Solution Book; the Book is versioned alongside the Solution.

9.1 Solution Book Structure

Book Element	Description
Charter	The foundational document: purpose, scope, goals, success criteria, and owner sign-off
Notes	Running log of decisions, observations, meeting notes, and ad hoc records
Parts	The canonical parts inventory: all Solution Parts referenced by this Solution
Vendors	Vendor records and qualification status for all suppliers used by this Solution
Binders	Logical groupings of related documents, assets, and content within the Book
Directories	Navigational indexes for all files, documents, and content in the Book
Lists	General-purpose ordered or unordered lists: requirements, features, risks, decisions
Collections	Named sets of related items: asset collections, reference sets, content libraries
Schedules	Planned timeline of activities, milestones, and deliverable dates
Timelines	Visual chronological representations of Solution history and planned events
Work Packages	Bounded units of work with defined scope, owner, deliverable, and effort estimate
Work Breakdown Structures (WBSs)	Hierarchical decomposition of all work required to deliver the Solution
Playbooks	Step-by-step guides for repeatable processes: release, incident response, onboarding
Roadmaps	Forward-looking plans: feature roadmaps, version plans, and strategic milestones

10. Solution Package

A Solution Package is the bundled, versioned, distributable form of a Solution. It encapsulates the Solution artifact, its metadata, configuration manifests, dependencies, and installation/deployment instructions into a single distributable unit that can be stored, transferred, deployed, or published.

10.1 Package Structure

Package Element	Description
Package ID	Unique identifier for this specific package version
Solution Reference	The Solution and Solution version this package represents
Package Version	The distribution version of this package (may differ from Solution version)
Artifact Manifest	List of all artifacts, binaries, containers, and assets included
Dependency Manifest	All declared dependencies with pinned versions and verification hashes
Configuration Template	Default configuration values and required parameter definitions
Installation / Deployment Instructions	Steps to deploy or install this package into a target environment
Build Attestation	Signed SLSA provenance record linking this package to its verified build inputs
License	Applicable license(s) governing use and distribution of this package
Changelog	Human-readable record of changes from the previous package version

11. Solution Compositions: Chain, Set, Kit & Tool Solution

Higher-order Solution constructs allow multiple Solutions to be grouped, linked, or composed into coherent offerings. These constructs operate at the Solution Registry and Factory Portfolio level.

Construct	Description	Typical Use
Solution Chain	An ordered, dependency-linked sequence of Solutions where the output of one feeds the next	End-to-end value chains; pipeline compositions; platform stacks
Solution Set	An unordered collection of Solutions that are deployed or managed together without strict dependency ordering	Product suites; feature bundles; release groups
Solution Kit	A curated, pre-configured bundle of Solutions, tools, and configurations designed to solve a specific class of problem	Starter kits; reference implementations; accelerators
Tool Solution	A Solution specifically classified as a tool — consumed by other Solutions or by SDEs as a tooling dependency	Compilers, linters, build systems, test runners, CLI utilities

12. Solution Orchestration

Solution Orchestration defines how the work of creating, delivering, and operating a Solution is organized into structured, traceable workflows and tasks. Orchestration connects Solution Models to Solution Factories, coordinating the sequence and parallelism of all factory activities.

12.1 Orchestration Hierarchy

Level	Description
Solution Orchestration	Top-level coordination model for all work associated with a Solution
Workflow	A named, directed graph of Tasks representing a complete end-to-end process
Task	The atomic unit of orchestrated work: an action with defined inputs, outputs, assignee, and completion criteria

12.2 Workflow Types

Workflow Type	Description
Build Workflow	Executes the hermetic build pipeline for a Solution or Component
Test Workflow	Runs a defined test suite against a Solution build
Release Workflow	Orchestrates build → test → stage → approve → deploy for a Solution release
Deployment Workflow	Executes the deployment of a Solution artifact to a target environment
Rollback Workflow	Restores a previous Solution version in a target environment
Provisioning Workflow	Creates and configures a new SDE from a template
Governance Workflow	Routes a Solution through review, approval, and CM promotion gates
Data Pipeline Workflow	Executes extract, transform, and load operations for Solution data

12.3 Task Schema

Task Field	Description
Task ID	Unique identifier
Task Name	Human-readable label
Workflow Reference	Parent workflow this Task belongs to
Task Type	build test deploy review approve notify script manual
Inputs	Data, artifacts, or conditions required to begin this Task
Outputs	Artifacts, state changes, or signals produced on completion
Assignee	User, team, service, or AI agent responsible for execution
Status	pending running succeeded failed skipped blocked
Dependencies	List of Task IDs that must complete before this Task may begin
Timeout	Maximum allowed execution time before the Task is marked as failed

Task Field	Description
Retry Policy	Number of retries and backoff strategy on failure

13. Solution Vendor

A Solution Vendor represents any external or internal entity that supplies components, parts, services, or capabilities to a Solution or Solution Factory. The Vendor Registry maintains a catalog of all approved suppliers and their associated Parts and Components.

13.1 Vendor Record

Vendor Field	Description
Vendor ID	Unique identifier for this vendor in the qala Vendor Registry
Vendor Name	Legal or trade name of the supplier
Vendor Type	Hardware Supplier Software Library SaaS Provider Service Provider Internal Team
Contact Information	Primary contact person, email, and communication channels
Parts Catalog	List of Parts and Components this vendor supplies, with part numbers and versions
Qualification Status	Approved Provisional Deprecated Blocked
SLA / Agreement Reference	Link to the governing contract, license, or SLA document
Lead Time	Typical delivery or provisioning time for this vendor's offerings

13.2 Vendor Qualification Workflow

- Submission: vendor submits Parts catalog, documentation, and certification evidence
- Review: governance team validates compliance with quality and security standards
- Provisional approval: vendor may be used in SANDBOX through TEST stages
- Full approval: vendor promoted to Approved status; Parts may be used in CM-stage Solutions
- Ongoing monitoring: SLA compliance, security advisories, and license changes tracked continuously

14. Solution Tooling

Solution Tooling defines the full hierarchy of tools available within a Solution Development Environment. Tooling is version-controlled, vendor-managed, and composable — from individual atomic tools up through toolsets, toolkits, toolchains, and tool suites.

14.1 Tooling Hierarchy

Level	Description
Tool	The atomic unit: a single executable, library, or service performing a specific function (e.g., compiler, linter)
Toolset	A named collection of related Tools serving a common purpose (e.g., 'Go Build Toolset': go, gofmt, golint, gotest)
Toolkit	A curated collection of Toolsets covering a full domain of practice (e.g., 'Backend Development Toolkit')
Toolchain	A linked, ordered sequence of Tools: each tool's output feeds the next (e.g., compile → link → sign → package)
Tool Suite	A governed, platform-approved set of Toolkits for a given Solution type or organizational standard

14.2 Tool Record

Tool Field	Description
Tool ID	Unique identifier in the qala Tool Registry
Tool Name	Human-readable name (e.g., rustc, docker, terraform)
Tool Version	Pinned semantic version used within this SDE
Tool Type	compiler linter formatter build-system test-runner container IaC IDE VCS scanner other
Vendor / Source	Tool publisher or open-source project origin
License	License type (MIT, Apache 2.0, proprietary, etc.)
Toolset Membership	Which Toolsets include this Tool
Toolchain Position	Position in any Toolchains this Tool participates in
Verification Hash	Cryptographic hash of the tool binary for supply-chain verification

15. Solution Registry

The Solution Registry is the authoritative catalog for all Solutions, Solution Models, Vendors, Tools, and Factories managed on the qala platform. It provides discovery, lineage tracking, governance assignment, and cross-reference services across the entire platform.

15.1 Registry Catalogs

Registry Catalog	Contents
Solution Registry	All Solutions: type, version, maturity, owner, SDE, factory, feature list
Solution Models Registry	All Blueprints, Designs, Architectures, Mockups, and Prototypes; linked to parent Solutions
Vendor Registry	All approved Vendors: parts catalog, qualification status, SLA references
Tool Registry	All Tools: version, type, license, verification hash, toolset membership
Factory Registry	All Solution Factories: SDE network, governance config, produced Solutions
Artifact Registry	All build artifacts: binary, version, attestation record, storage location
Portfolio Registry	All Solution Portfolios: composition, owner, factory association, maturity distribution

16. Solution Portfolio

A Solution Portfolio is a governed collection of Solutions managed under a common Solution Factory or organizational unit. Portfolios provide aggregate visibility into solution health, maturity distribution, value delivery, and strategic alignment.

16.1 Portfolio Record

Portfolio Field	Description
Portfolio ID	Unique identifier
Portfolio Name	Human-readable name (e.g., 'Platform Engineering Portfolio')
Owner	The organizational unit or governance board accountable for this Portfolio
Factory Reference	The Solution Factory that produces the Solutions in this Portfolio
Solutions	The set of Solutions contained in this Portfolio, with maturity and version tracking
Maturity Distribution	Aggregate view of how Solutions are distributed across maturity stages
Value Chain Coverage	Which value chain stages are actively instrumented for this Portfolio
Governance Status	Overall governance health: open gates, policy violations, pending approvals

17. Solution Value Chain

The Solution Value Chain describes the end-to-end sequence of activities, inputs, and outputs that transforms a Solution concept into a delivered, operational Solution. qala makes the value chain explicit, traceable, and governable.

Stage	Key Activities	Output
Ideation	Problem definition, goal setting, feasibility assessment	Solution Charter, value proposition
Design	Blueprint authoring, prototype creation, design review	Approved Solution Model
Build	Hermetic build, unit tests, CI pipeline execution	Verified build artifact
Test	Full test suite execution, defect resolution, security scan	Test-qualified artifact
Release	CM board review, version tagging, change log, sign-off	CM-approved release
Deploy	Deployment workflow execution to target environment	Running Solution instance
Operate	Monitoring, observability, incident response, patching	Stable operational Solution
Retire	Deprecation notice, migration path, archival	Archived Solution record

18. Governance, Standards & Policies

Governance in qala ensures that all Solutions, Factories, and Environments operate within defined quality, security, and compliance boundaries. Governance is enforced through policies, standards, templates, playbooks, and lifecycle gate criteria.

18.1 Governance Instruments

Instrument	Description
Policy	A declarative rule constraining behavior at the platform, factory, or solution level
Standard	A defined approach or specification that Solutions must conform to (e.g., API versioning, naming convention)
Template	A pre-approved scaffold for common Solution types, SDE configurations, or workflow definitions
Playbook	A step-by-step guide for executing a defined process (e.g., Incident Response, Release)
Gate Criteria	The specific conditions that must be satisfied for a Solution to advance between maturity stages
Audit Log	Immutable, tamper-evident record of all actions taken on Solutions, SDEs, and Factories
Ownership Record	Documented assignment of accountability for every Solution, Component, and SDE

18.2 Maturity Gate Criteria

Maturity Promotion	Required Gate Criteria
SANDBOX → DEV	Solution record created in Registry; SDE provisioned; owner assigned
DEV → NIGHTLY	CI pipeline configured; build succeeds; basic unit tests passing
NIGHTLY → TEST	All unit and integration tests passing on main branch; SAST scan clean; no P1 defects open
TEST → CM	All test suites passing; DAST scan clean; performance benchmarks met; governance board sign-off; build attestation generated

19. Solution Data Platform

The Solution Data Platform manages the full data lifecycle for the qala platform: metrics, events, analytics, master data, and AI-driven insights. Every data point is traceable to its origin Solution, SDE, or Factory.

19.1 Data Platform Capabilities

Capability	Description
Event Streaming	Platform-wide event bus (Apache Kafka) carrying Solution, SDE, Build, Security, and AI events
Metrics Collection	Time-series metrics from all Solutions, SDEs, and Factories: CPU, memory, latency, error rates
Log Aggregation	Centralized, indexed log collection from all platform services and Solution environments
Data Warehouse	Columnar analytics store for historical reporting, trend analysis, and AI model training
Master Data Management	Authoritative reference data for Solution types, status codes, and classification taxonomies
Analytics Dashboards	Pre-built dashboards for Solution health, factory throughput, defect density, and resource utilization
AI Inference	Real-time and batch inference endpoints for predictive capabilities throughout the platform
Data Lineage	Full provenance tracking: every data point traces to its origin event and transformation history

19.2 Event Topics

Event Topic	Published By
SOLUTION_EVENTS	Solution Registry — creation, update, maturity transitions
USER_EVENTS	User & Identity Service
SDE_EVENTS	SDE Management Service — provisioning, snapshot, rollback, lifecycle
CMS_EVENTS	Workspace & CMS Service — content create/update/delete
BUILD_EVENTS	Workflow & CI/CD Service — pipeline runs, build outcomes
ARTIFACT_EVENTS	Artifact & Package Management — uploads, downloads, deletions
DATA_EVENTS	Data Platform Service — pipeline runs, anomalies
SECURITY_EVENTS	Security & SEM Service — threats, policy violations, scans
NOTIFICATIONS	Notifications Service
AI_RECOMMENDATIONS	AI Agents Service

20. AI Integration

AI capabilities are embedded throughout the qala Solution System — not as an add-on layer, but as an integral part of every major subsystem. The AI Agents service provides intelligent hooks across the platform.

AI Capability	Subsystem	Description
SDE Optimization	SDE Management	Recommends configuration improvements for performance and cost efficiency
Pipeline Bottleneck Detection	CI/CD	Identifies slow stages and recommends parallelization or caching
Predictive Defect Detection	Testbed	Flags high-risk code areas using historical defect patterns
Test Case Generation	Testbed	Synthesizes test scenarios from specifications and past failures
Self-Healing Tests	Testbed	Automatically updates fragile tests when interfaces evolve
Anomaly Detection	Data Platform	Detects unusual patterns in metrics, logs, and security events
Resource Prediction	Data Platform	Forecasts compute and capacity needs based on activity trends
Security Threat Analysis	Security/SEM	Correlates security events to identify attack patterns and risks
Prototype Feedback Analysis	Solution Model	Synthesizes prototype results into design recommendations
Vendor Advisory Monitoring	Tooling	Monitors tool ecosystem for advisories and recommended upgrades
Backup Schedule Optimization	SDE Management	Recommends backup frequency based on change velocity and risk
Content Suggestions	Workspace/CMS	Recommends documentation structure and template selection

21. Security & Compliance

Security in qala is not a separate concern — it is embedded in the Solution lifecycle at every stage, enforced by the Security and Event Management (SEM) service and validated by automated scanning in every CI/CD pipeline.

Control	Description
RBAC	Role-Based Access Control enforced at every API endpoint and SDE operation
MFA	Multi-factor authentication required for all user and service accounts
mTLS	Mutual TLS for all service-to-service communication within the platform
Secrets Vault	HashiCorp Vault or cloud-native KMS for all credentials, certificates, and secrets
SAST	Static Application Security Testing on every commit
DAST	Dynamic Application Security Testing on every build targeting TEST+ maturity
Build Attestation	SLSA provenance records signed for every build output; verified before deployment
Supply Chain Verification	All tool and dependency binaries verified against cryptographic hashes before use
Immutable Audit Logs	All platform actions are recorded in tamper-evident, append-only audit logs
Data Encryption	Encryption at rest and in transit mandatory for all Solution and tenant data
Threat Detection	Real-time SEM service monitors for anomalous activity, policy violations, and attacks
SDE Quarantine	Compromised or suspicious SDEs can be automatically isolated by the SEM service

22. API Surface

qala exposes a versioned REST API surface. All endpoints route through the central API Gateway which enforces authentication, authorization, rate limiting, and observability.

22.1 Solution System Endpoints

Endpoint	Method(s)	Description
/solutions	GET, POST	List all Solutions or create a new Solution
/solutions/{id}	GET, PATCH, DELETE	Retrieve, update, or delete a Solution record
/solutions/{id}/components	GET, POST	List or add Components to a Solution
/solutions/{id}/components/{cid}	GET, PATCH, DELETE	Manage a specific Component
/solutions/{id}/components/{cid}/parts	GET, POST	List or add Parts to a Component
/solutions/{id}/model	GET, PUT	Retrieve or update the Solution Model
/solutions/{id}/features	GET, POST	List or add Features to the Solution features list
/solutions/{id}/maturity	GET, POST	Get maturity status or submit a maturity promotion request
/solutions/{id}/orchestration	GET, POST	View or create Workflows for a Solution
/solutions/{id}/testbed	GET	View testbed configuration and results
/solutions/{id}/value-chain	GET	View the value chain trace for a Solution
/solutions/{id}/book	GET, PUT	Retrieve or update the Solution Book
/solutions/{id}/package	GET, POST	Retrieve or create a Solution Package
/vendors	GET, POST	List or register Vendors
/vendors/{id}	GET, PATCH, DELETE	Manage a specific Vendor record
/tools	GET, POST	List or register Tools in the Tool Registry
/factories	GET, POST	List or create Solution Factories
/factories/{id}/sdes	GET, POST	List or create SDEs in a Factory
/factories/{id}/portfolio	GET	View the Solution Portfolio for a Factory
/sdes/{id}/snapshot	POST	Snapshot an SDE
/sdes/{id}/rollback	POST	Rollback an SDE to a prior snapshot
/sdes/{id}/clone	POST	Clone an SDE

Appendix A — Glossary

Term	Definition
Solution	The central root element of qala: a typed, versioned answer to a problem or goal, composed of Components and governed by a lifecycle
SDE	Solution Development Environment — the atomic operational unit providing all tools, configuration, and resources for creating a Solution
SF	Solution Factory — a coordinated collection of networked SDEs producing Solutions in a consistent, repeatable way
SFOS	Solution Factory Operating System — the qala platform itself
Solution Book	The complete knowledge and documentation repository for a Solution: charter, notes, schedules, WBSs, playbooks, and all reference content
Solution Package	A bundled, versioned, distributable unit containing a Solution artifact, its manifests, dependencies, and deployment instructions
Solution Chain	An ordered, dependency-linked sequence of Solutions forming an end-to-end value chain
Solution Set	An unordered collection of Solutions deployed or managed together
Solution Kit	A curated, pre-configured bundle of Solutions, tools, and configurations targeting a specific class of problem
Tool Solution	A Solution classified as a tool, consumed by other Solutions or SDEs as a tooling dependency
Solution Portfolio	A governed collection of Solutions managed under a common Factory or organizational unit
CM	Configuration Management — governed control of all environment and system configuration; highest maturity stage
CI	Continuous Integration — automated build and test on every code change
CD	Continuous Deployment/Delivery — automated promotion of validated artifacts to target environments
SAST	Static Application Security Testing — code analysis without execution
DAST	Dynamic Application Security Testing — testing against a running application
RBAC	Role-Based Access Control — access permissions determined by assigned roles
SLSA	Supply chain Levels for Software Artifacts — a security framework for build attestation and provenance
Toolchain	A linked, ordered sequence of Tools where each tool's output feeds the next
Blueprint	The formal structural specification of a Solution Model
Maturity	The lifecycle classification of a Solution: SANDBOX, DEV, NIGHTLY, TEST, or CM
Hermetic Build	A fully isolated, reproducible build environment with all dependencies frozen
Value Chain	The end-to-end sequence of activities transforming a Solution concept into an operational Solution
WBS	Work Breakdown Structure — hierarchical decomposition of all work required to deliver a Solution